



## **DNS AND WEB SERVER LINUX CONFIGURATION**

Group 10:

|                             |              |
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2CS1

**CEP CCIT FACULTY OF ENGINEERING**

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## PROJECT INFORMATION

Project Title : DNS and Web Server Linux  
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## **CERTIFICATE**

This is to certify that the project report titled "DNS and Web Server Linux Configuration" is an original work completed by Azzahra Fatma Wahida, Hilwa Annisa, and Nazilatun Natasya Ramadhani. This project has been submitted in partial fulfillment of their course requirement at the National Institute of Information Technology (NIIT).

The project report has been prepared under our guidance and supervision, and it is ensured that the work presented in this report is the result of the individual efforts of the aforementioned students. The contents of this report have not been submitted to any other institution or organization for the award of any degree, diploma, or other similar recognition.

Author acknowledge that the ideas, designs, and implementations presented in this project report are the intellectual properties of the students mentioned above. Any use or reproduction of this work must give proper credit to the original authors.

Author hereby endorse the authenticity and originality of the work presented in this project report and confirm that it meets the academic standards and requirements set forth by the National Institute of Information Technology (NIIT).

Coordinator:

Mr. Ivan Firdaus S.T

## **ACKNOWLEDGEMENT**

The author gratefully acknowledges the completion of this in-depth paper entitled “Configuring DNS and Linux Web Servers”. This paper comprehensively discusses how to configure DNS and Web Servers in Linux software.

The content of this paper provides a detailed overview of how to configure a DNS Server to translate human-readable domain names (e.g., google.com) into numeric IP address (e.g., 172.217.160.142) that computers use to communicate with each other and a Web Server to serve requests from users via HTTP or HTTPS protocols to access web pages. This paper is one way to understand Linux server as one of the courses in this second semester.

Depok, 25 March 2025

# SYSTEM ANALYSIS

## **System Summary:**

In the world of computer networks, servers have an important role in providing services to clients. One of the main services used is DNS Server (Domain Name System) and Web Server. DNS Server functions to translate IP Addresses into host names to make them more efficient and easy to remember. While the Web Server functions to serve requests from users via the HTTP or HTTPS protocol in order to access web pages. Along with technological developments, the need for a reliable and efficient server is increasing.

However, there are still many obstacles that are often faced in configuring DNS and Web Servers, such as errors in writing configuration files, failure in domain resolution, and lack of understanding in server security. If not configured properly, this can cause disrupted website access, increase the risk of security attacks, and hinder communication on the network.

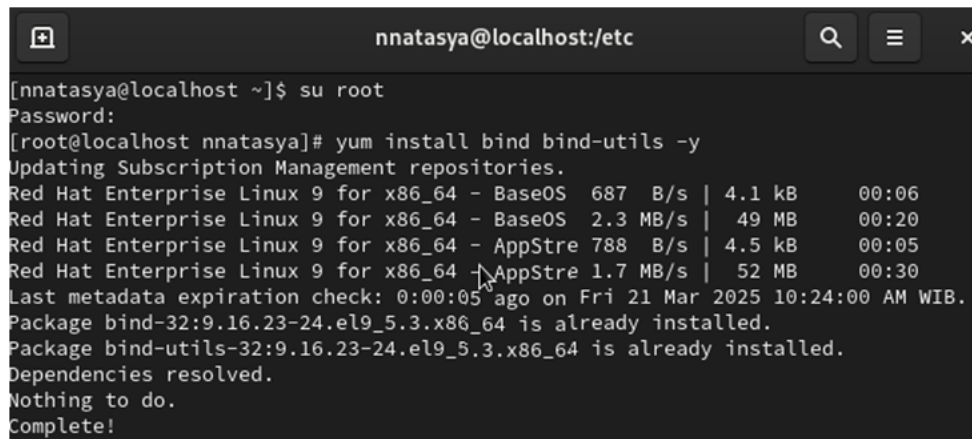
To address these issues, this paper aims to provide an in-depth understanding and guide to configuring DNS Server and Web Server in Linux. By discussing the steps of installation, configuration, and troubleshooting.

# INSTALLING COMPONENT DNS SERVER AND CONFIGURING DNS SERVER

1. Installing DNS Server (BIND9) in RHEL. Before installing DNS Server on RHEL, log in as root with the command

```
[nnatasya@localhost ~]$ su root
Password:
```

2. After that, use the BIND9 package for installation. Here are the steps:



```
nnatasya@localhost:/etc
[nnatasya@localhost ~]$ su root
Password:
[root@localhost nnatasya]# yum install bind bind-utils -y
Updating Subscription Management repositories.
Red Hat Enterprise Linux 9 for x86_64 - BaseOS 687 B/s | 4.1 kB 00:06
Red Hat Enterprise Linux 9 for x86_64 - BaseOS 2.3 MB/s | 49 MB 00:20
Red Hat Enterprise Linux 9 for x86_64 - AppStre 788 B/s | 4.5 kB 00:05
Red Hat Enterprise Linux 9 for x86_64 - AppStre 1.7 MB/s | 52 MB 00:30
Last metadata expiration check: 0:00:05 ago on Fri 21 Mar 2025 10:24:00 AM WIB.
Package bind-32:9.16.23-24.el9_5.3.x86_64 is already installed.
Package bind-utils-32:9.16.23-24.el9_5.3.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
```

3. After the installation is complete, configure the main named.conf file located in /etc/named/.conf.

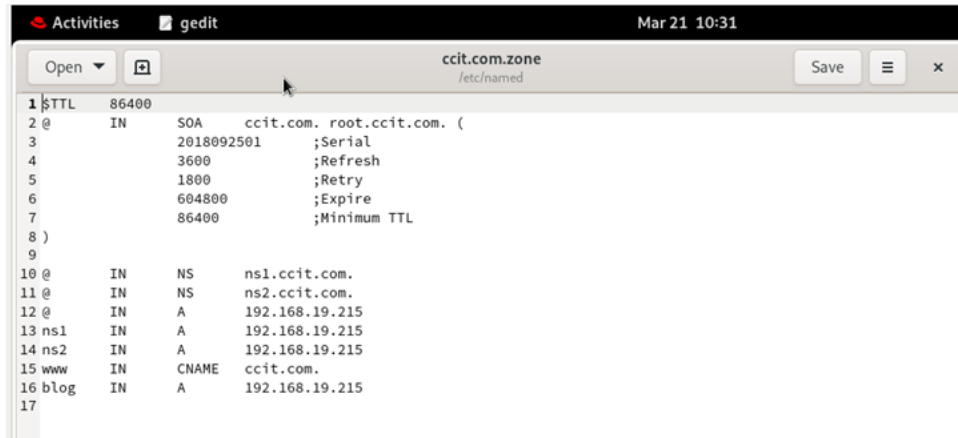
```
[root@localhost nnatasya]# cd /etc/
[root@localhost etc]# ls named
ccit.com.rev ccit.com.zone
```

4. Configuration Zone Forward. Zone forward is used to translate domain names into IP addresses. Add the following configuration in /etc/named.conf.

```
[root@localhost etc]# sudo gedit /etc/named/ccit.com.zone
```

# INSTALLING COMPONENT DNS SERVER AND CONFIGURING DNS SERVER

5. Then create a zone file in /etc/named/ccit.com.zone



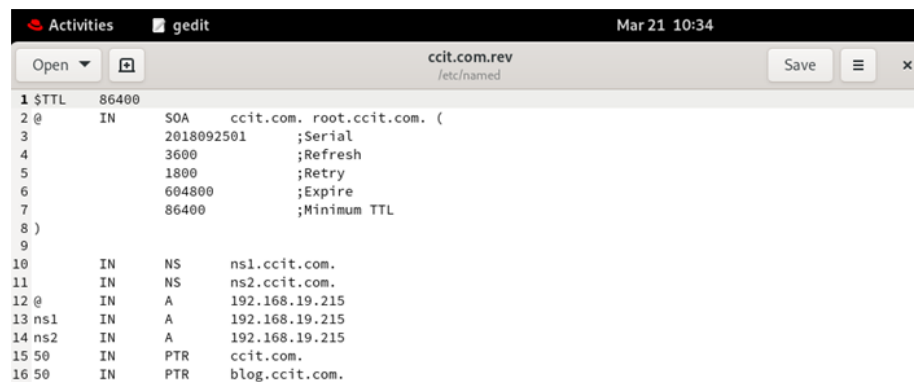
The screenshot shows a gedit window titled 'ccit.com.zone' with the following content:

```
1 $TTL 86400
2 @      IN      SOA      ccit.com. root.ccit.com. (
3          2018092501      ;Serial
4          3600             ;Refresh
5          1800             ;Retry
6          604800           ;Expire
7          86400            ;Minimum TTL
8 )
9
10 @      IN      NS       ns1.ccit.com.
11 @      IN      NS       ns2.ccit.com.
12 @      IN      A        192.168.19.215
13 ns1    IN      A        192.168.19.215
14 ns2    IN      A        192.168.19.215
15 www    IN      CNAME    ccit.com.
16 blog   IN      A        192.168.19.215
17
```

6. Configuration Zone Reverse. Zone reverse is used to translate IP addresses into domain names. Add the following configuration:

```
[root@localhost etc]# sudo gedit /etc/named/ccit.com.rev
```

7. And create a zone file in /var/named/ccit.com.rev



The screenshot shows a gedit window titled 'ccit.com.rev' with the following content:

```
1 $TTL 86400
2 @      IN      SOA      ccit.com. root.ccit.com. (
3          2018092501      ;Serial
4          3600             ;Refresh
5          1800             ;Retry
6          604800           ;Expire
7          86400            ;Minimum TTL
8 )
9
10 IN      NS       ns1.ccit.com.
11 IN      NS       ns2.ccit.com.
12 @      IN      A        192.168.19.215
13 ns1    IN      A        192.168.19.215
14 ns2    IN      A        192.168.19.215
15 50     IN      PTR      ccit.com.
16 50     IN      PTR      blog.ccit.com.
```

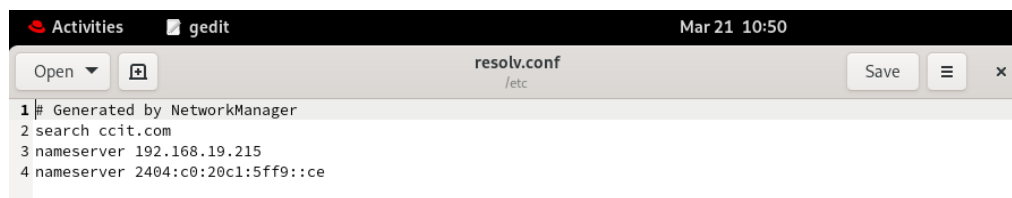
# INSTALLING COMPONENT DNS SERVER AND CONFIGURING DNS SERVER

- Restart Service and Configuration DNS Resolver. Once the configuration is complete, enable the DNS Server service:

```
[root@localhost etc]# systemctl enable named
```

```
[root@localhost etc]# systemctl restart named
```

```
[root@localhost etc]# sudo gedit /etc/resolv.conf
```



- Open Firewall. To make the DNS Server accessible, open the firewall

```
[root@localhost etc]# firewall-cmd --add-service=http --permanent  
Warning: ALREADY_ENABLED: http  
success
```

```
[root@localhost etc]# firewall-cmd --add-service=dns --permanent  
success
```

```
[root@localhost etc]# firewall-cmd --reload  
success
```

# INSTALLING COMPONENT DNS SERVER AND CONFIGURING DNS SERVER

10. Testing DNS Server. Use the following command to test if DNS is working:

```
[root@localhost etc]# nslookup ccit.com
Server:          192.168.19.215
Address:         192.168.19.215#53

Name:   ccit.com
Address: 192.168.19.215
```

```
[root@localhost etc]# dig ccit.com

;<<>> DiG 9.16.23-RH <<>> ccit.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 49175
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 1b51e99c0d69189a0100000067dce34dcfd764166a27f525 (good)
;; QUESTION SECTION:
;ccit.com.                IN      A

;; ANSWER SECTION:
ccit.com.                 86400   IN      A      192.168.19.215

;; Query time: 0 msec
;; SERVER: 192.168.19.215#53(192.168.19.215)
;; WHEN: Fri Mar 21 10:55:57 WIB 2025
;; MSG SIZE rcvd: 81
```

```
[root@localhost etc]# ping -c 4 ccit.com
PING ccit.com (192.168.19.215) 56(84) bytes of data.
64 bytes from localhost.localdomain (192.168.19.215): icmp_seq=1 ttl=64 time=0.034 ms
64 bytes from localhost.localdomain (192.168.19.215): icmp_seq=2 ttl=64 time=0.349 ms
64 bytes from localhost.localdomain (192.168.19.215): icmp_seq=3 ttl=64 time=0.098 ms
64 bytes from localhost.localdomain (192.168.19.215): icmp_seq=4 ttl=64 time=0.279 ms

--- ccit.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3050ms
rtt min/avg/max/mdev = 0.034/0.190/0.349/0.128 ms
```

# INSTALLING COMPONENT WEB SERVER AND CONFIGURING WEB SERVER

1. Apache Web Server Installation. Apache can be installed with the following command:

```
[root@localhost etc]# firewall-cmd --add-service=http --permanent
Warning: ALREADY_ENABLED: http
success
```

```
[root@localhost etc]# firewall-cmd --reload
success
```

2. Service HTTPD. systemctl start httpd to start the httpd service. systemctl status httpd to view httpd active or inactive status.

```
[root@localhost nnatasya]# sudo systemctl status httpd
Unknown command verb status.
[root@localhost nnatasya]# sudo systemctl status httpd
* httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Mon 2025-03-24 15:42:45 WIB; 4min 3s ago
     Docs: man:httpd.service(8)
  Main PID: 922 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 11768)
   Memory: 49.9M
      CPU: 223ms
  CGroup: /system.slice/httpd.service
          └─922 /usr/sbin/httpd -DFOREGROUND
            └─944 /usr/sbin/httpd -DFOREGROUND
              └─946 /usr/sbin/httpd -DFOREGROUND
                └─947 /usr/sbin/httpd -DFOREGROUND
                  └─948 /usr/sbin/httpd -DFOREGROUND

Mar 24 15:42:45 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Mar 24 15:42:45 localhost.localdomain httpd[922]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using localhost.localdomain
Mar 24 15:42:45 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
Mar 24 15:42:45 localhost.localdomain httpd[922]: Server configured, listening on: port 80
lines 1-20/20 (END)...skipping...
* httpd.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; enabled; preset: disabled)
   Active: active (running) since Mon 2025-03-24 15:42:45 WIB; 4min 3s ago
     Docs: man:httpd.service(8)
  Main PID: 922 (httpd)
    Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 177 (limit: 11768)
   Memory: 49.9M
      CPU: 223ms
  CGroup: /system.slice/httpd.service
          └─922 /usr/sbin/httpd -DFOREGROUND
            └─944 /usr/sbin/httpd -DFOREGROUND
              └─946 /usr/sbin/httpd -DFOREGROUND
                └─947 /usr/sbin/httpd -DFOREGROUND
                  └─948 /usr/sbin/httpd -DFOREGROUND

Mar 24 15:42:45 localhost.localdomain systemd[1]: Starting The Apache HTTP Server...
Mar 24 15:42:45 localhost.localdomain httpd[922]: AH00558: httpd: Could not reliably determine the server's fully qualified domain name, using localhost.localdomain
Mar 24 15:42:45 localhost.localdomain systemd[1]: Started The Apache HTTP Server.
Mar 24 15:42:45 localhost.localdomain httpd[922]: Server configured, listening on: port 80
```

# INSTALLING COMPONENT WEB SERVER AND CONFIGURING WEB SERVER

3. Ifconfig. Check the ip address with ifconfig, to be able to access the browser, whether the web server is running or not. Relationship with DNS server that can be accessed by domain name

```
[root@localhost etc]# ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.19.215 netmask 255.255.255.0 broadcast 192.168.19.255
    inet6 fe80::a00:27ff:fe5c:5def prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:5c:5d:ef txqueuelen 1000 (Ethernet)
    RX packets 583 bytes 113511 (110.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 390 bytes 47901 (46.7 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 299 bytes 29613 (28.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 299 bytes 29613 (28.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

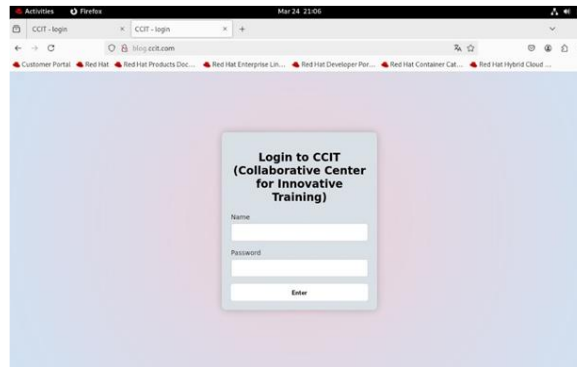
4. Apache Web Content. The location of the web html file is in /usr/share/httpd/noindex/index.html

```
[root@localhost nnatasya]# gedit /usr/share/httpd/noindex/index.html
```

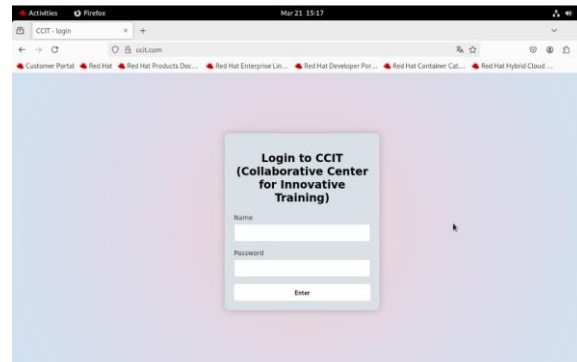
```
1<!DOCTYPE html>
2<html lang="id">
3<head>
4    <meta charset="UTF-8">
5    <meta name="viewport" content="width=device-width, initial-scale=1.0">
6    <title>CCIT - login</title>
7    <style>
8        body {
9            font-family: 'Poppins', sans-serif;
10           margin: 0;
11           padding: 0;
12           background: radial-gradient(circle, #edd9d9, #d0e1f2);
13           display: flex;
14           justify-content: center;
15           align-items: center;
16           height: 100vh;
17       }
18       .login-container {
19           background: #dae1e6;
20           padding: 20px;
21           border-radius: 10px;
22           box-shadow: 0px 0px 15px rgba(0, 0, 0, 0.2);
23           width: 300px;
24           text-align: center;
25       }
26       .login-container h2 {
27           color: black;
28           font-weight: 600;
29       }
30       .input-group {
31           margin: 15px 0;
32           text-align: left;
33       }
34       .input-group label {
35           display: blok;
36           font-size: 14px;
37           color: #333;
38           font-weight: 400;
39       }
40       .input-group input {
41           width: 100%;
```

# INSTALLING COMPONENT WEB SERVER AND CONFIGURING WEB SERVER

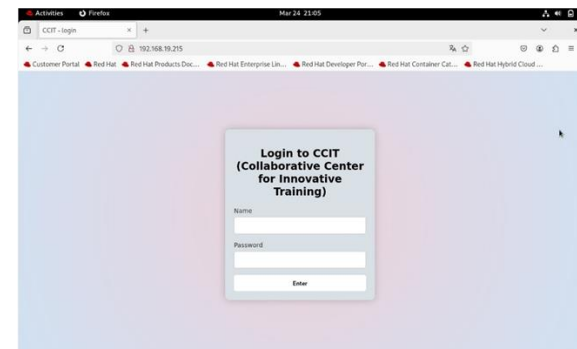
5. Log in to the website via [blog.ccit.com](http://blog.ccit.com)



6. Log in to the website via [ccit.com](http://ccit.com)



7. Log in to the website via IP Address



# REQUIREMENTS

Hardware : HP

Operating System : Microsoft Windows 10, Linux RHEL 9.5

Software : 1. Virtual Box

2. Red Hat Enterprise Linux

3. Microsoft word

4. Microsoft PowerPoint

## PROJECT FILE DETAILS

| No | Filename                                      | Remarks                                                     |
|----|-----------------------------------------------|-------------------------------------------------------------|
| 1  | DNS and Web Server<br>Linux configuration.pdf | Microsoft Words contain<br>research paper about the project |
| 2  | Project 2<br>Presentation.pptx                | Presentation file                                           |